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Bihar Engineering University, Patna

B.Tech 2nd Semester Exam., 2021

Subject: MATHEMATICS—II (Differential Equations)

Code: 101202 | ***Time: 3 hours*** | ***Full Marks: 70***

Instructions:-

(i) *The marks are indicated in the right-hand margin.*
(ii) *There are NINE questions in this paper.*
(iii) *Attempt FIVE questions in all.*
(iv) *Question No. 1 is compulsory.*

Q.1 Choose the correct answer (any seven): **[2×7=14]**

(a) Integrating factor of the differential equation $dy = (e^{x-y})(e^x - e^y)\,dx$ is

(i) e^{e^x} (iii) e^x
(ii) e (iv) e^{2x}

(b) The particular integral (PI) of the differential equation $d^2y/dx^2 + 4y = \cos 2x$ is

(i) $\sin 2x/2$ (iii) $[x\sin 2x] / [4]$
(ii) $[x\sin 2x] / [2]$ (iv) $[x\cos 2x] / [2]$

(c) The differential equation whose auxiliary equation has the roots 0, -1, -1 is

(i) $d^3y/dx^3 - 2d^2y/dx^2 + dy/dx = 0$ (iii) $d^2y/dx^2 + 2dy/dx + y = 0$
(ii) $d^3y/dx^3 + 2d^2y/dx^2 + dy/dx = 0$ (iv) $d^3y/dx^3 + 2d^2y/dx^2 + y = 0$

(d) If $J_0(x)$ and $J_1(x)$ are Bessel functions, then $J_1'(x)$ is given by

(i) $-J_0(x)$ (iii) $J_0(x) + 1/xJ_1(x)$
(ii) $J_0(x) - 1/xJ_1(x)$ (iv) $J_0(x)$

(e) The complete solution of the partial differential equation $p^2 + q^2 = 2$ is

(i) $z = ax + \sqrt{(2 - a^2)}y + c$ (iii) Either (i) or (ii) true
(ii) $z = ax - \sqrt{(2 - a^2)}y + c$ (iv) (i) and (ii) both true

(f) The partial differential equation $5[\partial^2 z] / [\partial x^2] - 5([\partial z] / [\partial x])^2 + 6[\partial^2 z] / [\partial y^2]x = xy$ is classified as

(i) parabolic (iii) elliptic
(ii) hyperbolic (iv) None of the above

(g) The solution of the partial differential equation $x^3[\partial u] / [\partial x] + y^2[\partial u] / [\partial y] = 0$ if $u(0, y) = 10\,e^{5/y}$ using method of separation of variable is

(i) $10\,e^{5/2x^2}e^{5/y}$ (iii) $10\,e^{-5/2y^2}e^{-5/x}$
(ii) $10\,e^{-5/2y^2}e^{5/x}$ (iv) $10\,e^{-5/2x^2}e^{5/y}$

(h) If $f(z) = u(x, y) + iv(x, y)$ is an analytic function, then $f'(z) =$

(i) $[\partial u] / [\partial x] - i[\partial v] / [\partial x]$ (iii) $[\partial v] / [\partial y] - i[\partial v] / [\partial x]$
(ii) $[\partial u] / [\partial x] - i[\partial u] / [\partial y]$ (iv) $[\partial u] / [\partial x] + i[\partial u] / [\partial y]$

(i) The value of $\int_C [3z^2 + 7z + 1] / [z + 1]dz$ where C is $|z| = 1$ is

(i) $2\pi i$ (iii) πi
(ii) 0 (iv) $\pi i/2$

(j) The value of $\Delta\tan^{-1}x$ is

(i) $\Delta\tan^{-1}([h] / [1 + hx + x^2])$ (iii) $\Delta\tan^{-1}([h - 2x] / [1 + hx + x^2])$
(ii) $\Delta\tan^{-1}([h + 2x] / [1 + hx + x^2])$ (iv) $\Delta\tan^{-1}([h + 2x] / [1 - hx - x^2])$

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Q.2 Solve both questions: **[7+7=14]**

(a) Solve: $2ydx + x(2\log x - y)dy = 0$

(b) Solve: $y - 2px = \tan^{-1}(xy^2)$

Q.3 Solve both questions: **[6+8=14]**

(a) Use the method of variation of parameters to find the solution of the given differential equation $y'' - 2y' + y = e^x \log x$

(b) Find the series solution of the differential equation $(x^2 - 2x + 1)d^2y/dx^2 + (4x - 4)dy/dx + (x^2 - 2x + 3)y = 0$ about the point $x = 1$.

Q.4 Solve both questions: **[6+8=14]**

(a) Prove: $\int J_3(x)\,dx = c - J_2(x) - 2/x J_1(x)$, where c is arbitrary constant.

(b) Prove: $\int_{-1}^{1} x^2 P_{n+1}(x)\, P_{n-1}(x)\,dx = [2n(n+1)] / [(2n-1)(2n+1)(2n+3)]$

Q.5 Solve both questions: **[7+7=14]**

(a) Solve: $x^2(y - z)\,p + y^2(z - x)\,q = z^2(x - y)$

(b) Find the complete integral of $(p^2 + q^2)y = zq$

Q.6 Solve both questions: **[8+6=14]**

(a) Prove that the general solution of the partial differential equation $(D - mD' - a)^2 z = 0$ where $D = [\partial] / [\partial x]$, $D' = [\partial] / [\partial y]$ is given as $z = e^{\alpha x} f_1(y + mx) + x e^{\alpha x} f_2(y + mx)$

(b) Find the general solution of the given partial differential equation $(D^3D'^2 + D^3D^2 - 5D^2D'^2 - 2D^3D' + 6D^2D')z = e^{x-y}$

Q.7 Solve this question: **[14]**

Find the D'Alembert's solution of the wave equation $[\partial^2 y] / [\partial t^2] = c^2 [\partial^2 y] / [\partial x^2]$

Q.8 Solve this question: **[14]**

Find the solution of the differential equation $[\partial u] / [\partial t] = k[\partial^2 u] / [\partial x^2]$ subject to conditions—

(i) u is not infinite at t to ∞

(ii) $[\partial u] / [\partial x] = 0$ for $x = 0$ and $x =$ ell

(iii) $u =$ ell$x - x^2$ for $t = 0$ between $x = 0$ and $x =$ ell.

Q.9 Solve this question: **[14]**

Describe the Runge-Kutta method of fourth order for the solution of initial value problem. Given the initial value problem $y' = 1 + y^2$, $y(0) = 0$. Find $y(0.6)$ by using Runge-Kutta method of fourth order by taking h = 0.2.

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